

Where Energy Recovery Stalls

Two margins and one crucial risk-set distinction

PAPER TITLE

Selective Entry and Structured Electricity-Recovery Performance in Japan's Waste-Incineration Fleet: A Facility-Level Panel Study

15-18 MINUTE ROUTE

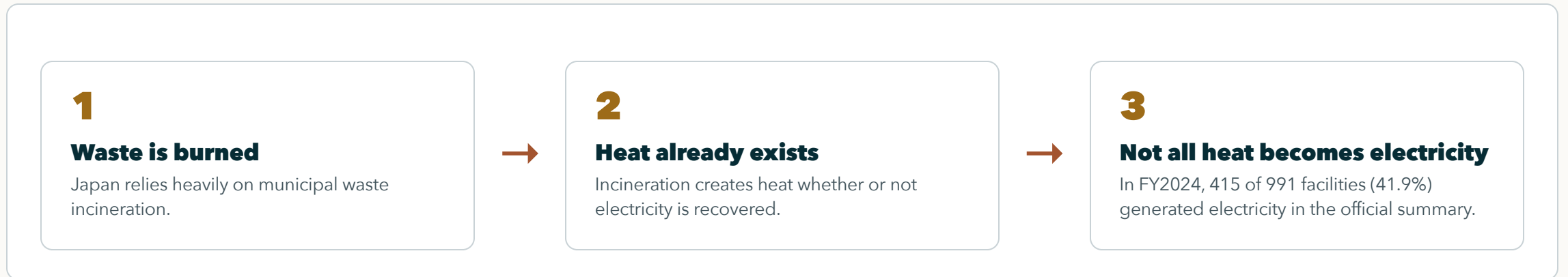
DATA + METHOD

RESULTS + LIMITS

Pann Phetra

Paper discussion briefing

Why This Matters



The same waste-treatment process can either recover useful power or miss that energy-recovery opportunity.

The Real Question

EXPECTED PART

Scale may matter, but the age story depends on which facilities count as candidates.

The paper tests broad asset entry, conversion of active plants, and performance after entry as distinct outcomes.

Two Questions, One Fleet

CORE LOGIC

One fleet average hides two different bottlenecks.

QUESTION 1

Which plants start generating electricity?

This is the entry bottleneck.

QUESTION 2

Among generators, who produces more electricity per tonne?

This is the performance bottleneck.

Entry = a facility first reports positive installed generation capacity.

MWh/t = electricity produced per tonne of waste processed.

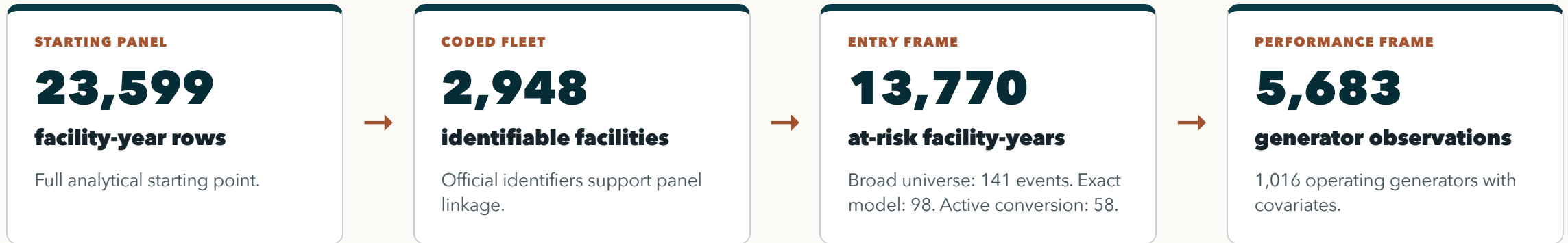
Bottleneck = the step where improvement is most constrained.

What the Data Can See

Observed layer	Observed fields	Analytical role
MOE General Waste Treatment Survey	National municipal waste-treatment facility records, FY2005-FY2024	Base facility-year panel and power-generation outcome
Facility identifiers and location	Official code, facility name, prefecture, fiscal year	Panel linkage, prefecture controls, clustered uncertainty, duplicate-code checks
Facility operating attributes	Installed capacity/output, throughput, design capacity, start year/age, furnace and operation type	Entry risk set and generator-output comparison
Context and plausibility controls	Heating value and reported generation efficiency	Comparability and engineering-oriented validation

The panel observes facility reporting patterns well. It does not directly observe internal retrofit contracts, municipal bargaining, or full lifecycle emissions.

How the Panel Becomes Two Frames



The design distinguishes broad asset entry, conversion among active non-generators, and performance after entry.

Why Two Samples Are Needed

Question	Sample frame	Outcome	What it tests
Who first reports installed capacity?	Facilities first observed without power generation	First report of electricity generation	Whether energy recovery diffuses into the lagging fleet
Who generates well?	Identifiable operating generators	MWh generated per tonne processed	Whether performance converges after entry

One model for everything would mix the gate into generation with performance after entry, hiding where the bottleneck actually sits.

Method: First Entry Into Generation

PLAIN MODEL

$\text{Pr}(\text{first installed capacity in year } t) = f(\text{prior age, prior capacity, year, elapsed time at risk})$

RISK SET

Only initial non-generators are candidates.

Facilities already generating in their first observed year are left-censored for first-entry analysis.

TIMING DISCIPLINE

Predictors are lagged.

Age and capacity are measured before first entry, not after the event.

TWO ESTIMANDS

Broad assets and active plants.

The active model requires positive throughput in the prior fiscal year.

Plain question: among facilities still outside power generation, who first reports generation in the next fiscal year?

Method: Output Among Generators

OUTCOME MODEL

$\log(\text{MWh per tonne}) = \text{age/vintage} + \text{capacity} + \text{utilization} + \text{heating value} + \text{year} + \text{technology}$

OUTCOME

Electricity recovered per tonne processed.

This measures output intensity, not just the existence of a generator.

PRIMARY MODEL

Year + technology OLS.

Furnace type, operating mode, facility type, and furnace count are observed controls.

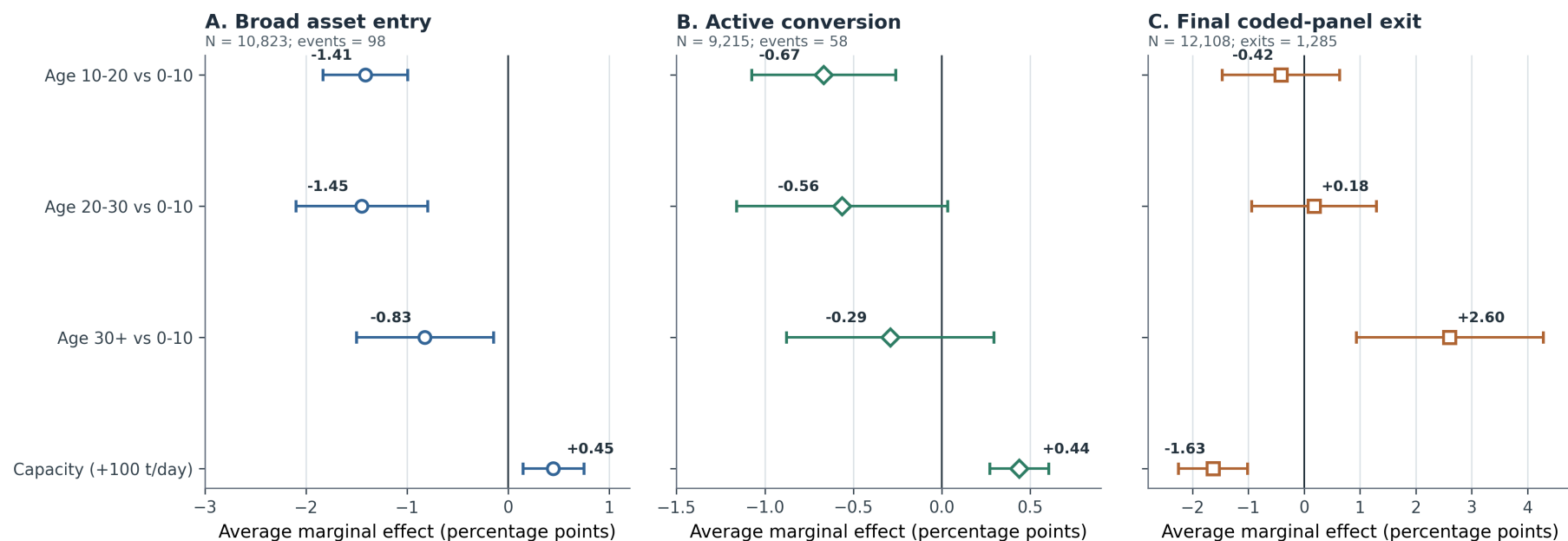
INTERPRETATION

This is diagnostic, not causal.

It compares generators within common years without claiming an intervention effect.

The coefficient is an age/vintage comparison across facilities, not the causal effect of making one plant older.

Result 1: Scale Is Robust; Age Depends on the Frame



Capacity predicts entry in both frames (+0.45 and +0.44 pp per 100 t/day); age differences shrink among active plants.

Interpretation: The Risk Set Changes the Story

BROAD ASSET ENTRY

98

exact-year events

Age AMEs: -1.41, -1.45, -0.83 pp.

ACTIVE CONVERSION

58

exact-year events

Age AMEs attenuate to -0.67, -0.56, -0.29 pp.

SCALE RESULT

+0.45 / +0.44

pp per 100 t/day

Positive and precisely estimated in both frames.

40 of 98 broad events follow zero or missing prior-year throughput. Commissioning and rebuild pathways materially affect the broad age result.

Entry Pathways: Modernization, Not One Mechanism

RESET / REBUILD-LIKE

50

observed events

Capital-side modernization is empirically present.

CONTINUITY / UPGRADE-LIKE

36

observed events

Some cases remain consistent with in-place upgrade.

AMBIGUOUS / OTHER

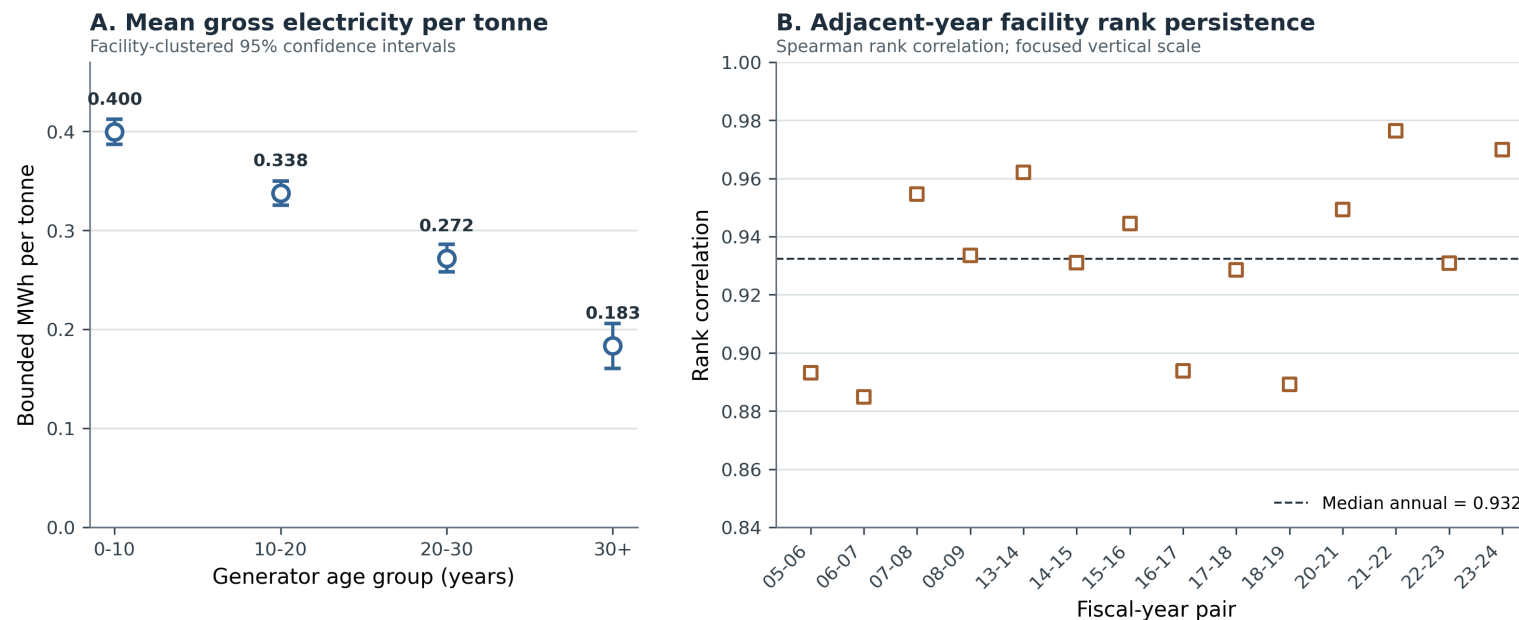
55

observed events

12 placeholder, 42 non-adjacent, and 1 unresolved event.

The audit bounds possible pathways, but it does not prove that replacement is the only mechanism.

Result 2: Generation Status Is Not Enough



Among generators, output intensity still differs sharply: age 0-10 averages 0.400 MWh/t, while age 30+ averages 0.183 MWh/t.

Interpretation: Ranks Persist Across Coded Years

FULL SAMPLE

0.1499

within-to-total variance ratio

Most generator-output variation is between facilities.

CODED WINDOWS

**0.1795 →
0.0956**

within-to-total ratio

FY2005-09 versus FY2013-24.

ADJACENT-YEAR RANKS

0.9325

pooled correlation

4,368 exact pairs across 915 facilities.

FY2010-FY2012 official codes are missing, so the paper uses FY2005-FY2009 and FY2013-FY2024 as coded windows, not a Fukushima treatment design.

Stress Tests: The Pattern Survives

Stress test	Why it matters	What survives
Composite facility-ID sensitivity	Checks whether duplicate official codes drive the adoption result	Age penalties and positive capacity effect remain stable
Active-conversion risk set	Removes zero/missing prior-throughput pathways	Scale persists; the age gradient attenuates
Technology-adjusted primary model	Controls observed furnace and operating configuration	Age/vintage stays negative; scale and utilization stay positive
Engineering outcomes + lagged predictors	Checks outcome definition and same-year simultaneity	Thermal conversion, reported efficiency, and lagged models agree

These checks do not make the estimates causal. They show the diagnostic pattern is not a fragile artifact of one coding choice.

Data Limits: Disclosed and Tested

DUPLICATE-CODE CONCERN

39

official codes affected

444 source rows use codes that duplicate within at least one fiscal year.

MISSING-CODE CONCERN

907

operating-generator rows

Rows missing official codes are excluded from the canonical regression frame.

PLAUSIBILITY CONCERN

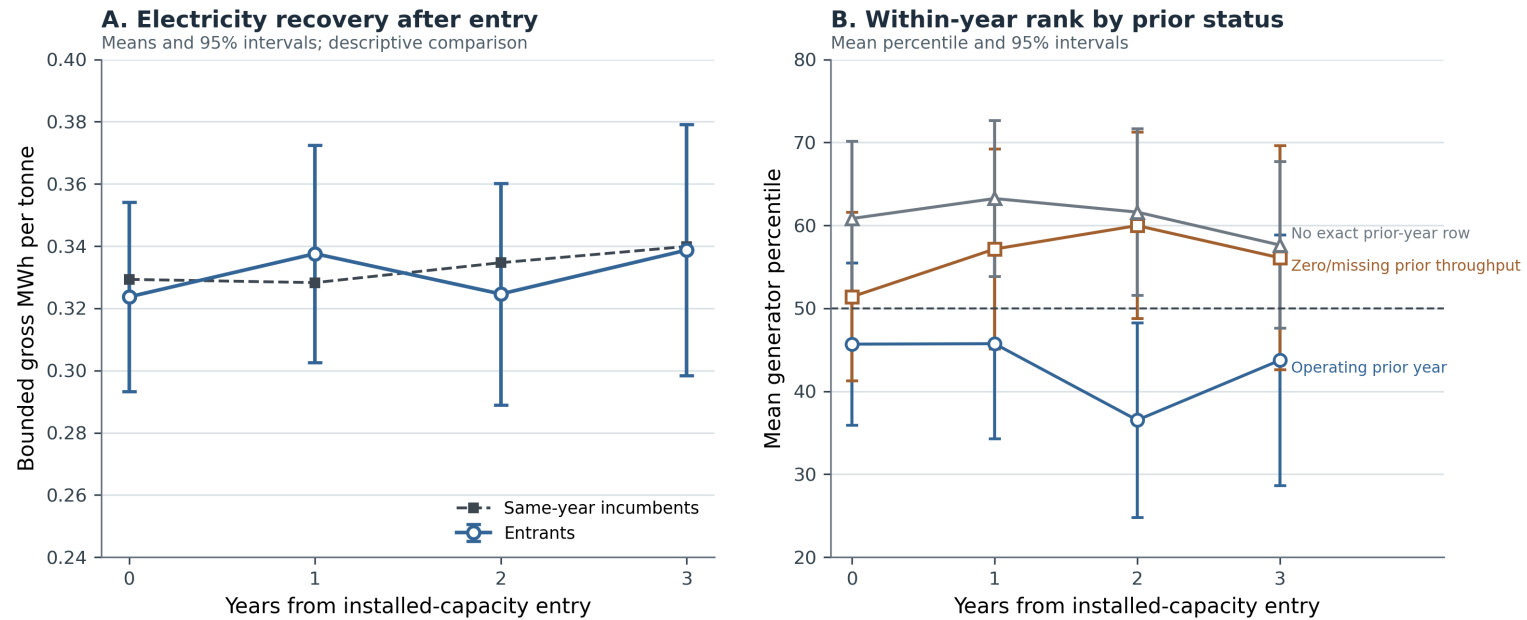
569

heating-value rows

Rows outside 3-25 MJ/kg are checked through sensitivity restrictions.

These limits are disclosed and stress-tested. The evidence maps bottlenecks; it does not claim a perfect engineering census or fully identified causal mechanism.

What Happens After Entry?



Entrants begin near the middle of the same-year generator distribution. Follow-up falls from 125 events at time zero to 71 at time three, so this is descriptive, not an entry effect.

Contribution: Two Margins, Clear Boundaries

WEAK VERSION

Newer and larger plants have advantages.

This is plausible but not enough for a paper by itself.

STRONGER VERSION

The result has clear boundaries.

Scale is robust, age depends on the risk set, and entry does not guarantee a frontier position.

Planning should distinguish broad asset entry, active conversion, and generator performance before choosing an intervention.

Next Step: Test Mechanisms

CAPITAL RENEWAL

Link entry to investment and rebuild histories.

This would test whether selective entry is mainly capital replacement or retrofit.

GOVERNANCE AND ROUTING

Link facilities to municipal decisions.

This could explain why some plants start generation or improve output.

CLIMATE AND COMPARISON

Add lifecycle or cross-fleet evidence.

This would move from energy-recovery diagnosis toward climate or international comparison.

The next research step is moving from diagnostic mapping to mechanism testing.

Discussion Questions

Three linked outcomes shape the story: asset entry, active conversion, and generator performance.

Is the three-outcome framing convincing, and is the broad-versus-active entry distinction clear enough? Which mechanism should future work test first?